

Dr. Leyna May-Gurr

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SUMMARY

Software engineer with a Mathematics PhD, currently building full-stack features at a women's fitness startup. A confident communicator through running seminar series, organising international conferences, and Women in Tech workshops. Based in Munich and eligible to work in Germany without visa sponsorship.

EDUCATION

University of Sheffield	2025
PhD in Mathematics. Thesis 'Hidden Symmetries, Trees and Operads'	UK
Durham University	2020
Masters in Mathematics, First class honours. Thesis 'Morse Theory: Morse-Bott Functions and Homology'	UK

EXPERIENCE

Software Engineer	2026 – Present
<i>Bobbi</i>	<i>Remote</i>
- Second developer on a two-person engineering team, working on creating a women's fitness tracking app - Building the workout guide pages in JavaScript and Vue.js	
Volunteer Tutor	2026 – Present
<i>ReDI School of Digital Integration</i>	<i>Munich</i>
Tutoring Intro to Coding, Data and Design (Python, HTML/CSS, UI/UX) to adult students at a non-profit digital school for newcomers to Germany	
Graduate teacher	2021 – 2025
<i>University of Sheffield</i>	<i>UK</i>
- Taught Python and R to undergraduate engineering students, led mathematics and engineering tutorials, and marked assessments with constructive feedback on student code - Supervised a UKRI-funded undergraduate summer project on knot theory (June–July 2023)	

PROJECTS & PUBLICATIONS

Ultimate Noughts and Crosses <i>Python, FastAPI, JavaScript, Vue.js, HTML, CSS</i>	2026
- Built a real-time multiplayer “ultimate” variant, nine nested boards, with a Vue.js front-end and a FastAPI backend handling WebSockets, game state, and room joining - Deployed to Oracle Cloud via GitHub Actions; playable at noughts-and-crosses.leynamay.com, source at github.com/LeynaM/noughts-and-crosses	
Tilt Game <i>JavaScript, Three.js, React, Node.js, SQLite, HTML, CSS</i>	January 2025
- Built a physics-based browser game in Three.js with custom algorithms for ball movement, platform tilting, and collision detection - Built mouse controls, stored scores in SQLite behind a React leaderboard, and deployed to Oracle Cloud via GitHub Actions; playable at tilt.leynamay.com, source at github.com/LeynaM/tilt-game	
Quartic graphs which are Bakry-Émery curvature sharp <i>Python</i>	2020
Designed a recursive algorithm to generate unique graphs with given curvature properties; published in Discrete Mathematics 343(3) (doi.org/10.1016/j.disc.2019.111767)	

SKILLS

Frontend: JavaScript, Vue.js, React, HTML, CSS

Backend: Python, FastAPI, Node.js, SQLite

Tools: Git, Linux, LaTeX | **Languages:** English (Native), German (B1)